Reflection on Behaviourism

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Understandings of Learning/Knowledge

- Learning is viewed as a change in observable behaviour driven by environmental conditions
    - **Stimulus** (trigger) →**response** (action)
    - e.g., "2+4=?" flashcard is the stimulus, saying "6" is the response
- **Reinforcement matters:** behaviour followed by reinforcement is more likely repeated
- Does not focus on brain processes, mental structures ("black box" mind)
    - No internal representations - focus is on observable and environmental
- **Teaching =** arranging the right reinforcement at the right time so behaviour becomes more likely
    - Occasion/Stimulus → Response/behaviour → consequence/Feedback
- **Shaping =** building complex skill/behaviour step by step by reinforcing small improvements
- **Knowledge** - having organized accumulation of associations between stimuli and response
    - Come across stimuli in particular env't and recall the response you had as a learner
    - Knowledge is the acquired behaviour itself or response in different settings as well?
    - It is not so developed, observed association between stimuli and response

Views on Memory

- Memory is not understood as "internal storage system"
    - More like a history of reinforcement - learned responses from previously reinforced behaviour - remembering of a habit
    - Do not study memory in the sense like other fields do
- Forgetting happens because a behaviour hasn't been used in a long time
    - The connection between situation and behaviour weakens, maybe situation does not happen so behaviour(response) does not occur and thus we forget how to respond to that situation
- **Extinction**: a previously reinforced/conditioned response decreases when no longer maintained/paired with the initial stimulus
    - E.g., maybe Little Albert's fear of rats could have been extinguished through exposure without the noise?
- **Vanishing =** technique where gradual removal of text/letters/words as external support (e.g., learning a poem by heart) as learner is able to give the response without them

Views on Transfer

- Transfer occurs through "**generalization**"
    - Applying a learned behaviour to a new situation sharing features with a previous one
    - e.g., a student who learns to identify elm trees can also identify maple trees using the same process, or Little Albert but with rabbit/other small furry animals

- Transfer happens when instruction/teaching provides practice across different related contexts and not just the same one
    - If learner does not practice across contexts and only in specific situation, transfer is limited/difficult

Basic Principles of Instructional Design

- Focuses on observable/measurable outcomes, pre-assessment, step-by-step mastery, and reinforcement
- Complex learning is broken into small steps:
    - Easy → slightly harder → harder → final desired behaviour
- **Immediate feedback**: reinforcement is more effective the closer it happens to the behaviour
    - Instructional tools should eliminate delay between response and feedback
- **Reinforcement over punishment:** correct behaviour should have a positive consequence, not fear/anxiety/threats
- **Teaching machines** are built on operant conditioning
    - Student answers and gets immediate feedback, relying on small steps

Formulating Instruction and Assessment

- Instruction is structured by providing:
    - A **stimulus** with temporary **cues**
    - **Repeated practice**
    - **Reinforcement** to link stimulus with behaviour
    - Via the above, the learned can eventually perform a specific action/behaviour when encountering a specific trigger/stimulus
    - e.g., a new manager given past agendas/templates as cues, practicing and receiving feedback, until they can eventually create an agenda from a verbal command alone
- **Behavioural chaining**: instruction is broken into a tight series of incremental steps
    - Learner masters each step to learn the full chain
    - E.g., teaching steps of washing hands
- **Programmed Instruction** breaks material into small steps/"frames"
    - Each frame asks the student to respond, then immediately shows if correct
    - Self-paced, moving to the next frame only once the current one is mastered
    - Should be lots of prompts and hints so student almost never answers incorrectly
- Grades mean something different
    - Show "how far you've gotten" through material, not "how much you understood" (e.g., a C means "halfway done," not "halfway understood").
- Goals should be written as observable behaviours, not vague concepts
    - e.g., "can label parts of a plant cell and explain what each does" vs "understand photosynthesis"
- Observable and measurable outcomes
    - Preassessment should happen

- Steps get more complex

Overall Reflection

- Behaviourism works best for teaching skills that have very clear goals and can be practiced over and over again
    - E.g., factual recall or procedural skills
- Due to the focus being on observable behaviour, it does not work well for explaining complex mental processes like reasoning or creativity
- Burden is on teacher if student does not learn, teacher did not set up right environment for learning to occur
    - Teacher is responsible for setting up correct stimulus and providing appropriate feedback
- Overall, it is helpful for understanding how feedback and reinforcement can help learning
- It can also be useful for designing specific types of learning if your goal is:
    - e.g.,memorization, sports (practicing shooting or passing or dribbling), routines/procedures etc.
- Examples of behaviourism being used as a tool in instructional design could be:
    - Anki
        - Create your own flashcards, test them against yourself and immediately shown right or wrong answer
        - You, yourself give feedback on how much you know and the system shows the card more or less based on your own feedback
    - Duolingo
        - Gives feedback immediately based on right or wrong answers
        - There are levels that need to be accomplished before moving forward
        - Weaves in things you may have been a bit weaker in, into newer lessons so you do not forget
    - Corporate Training/onboarding modules
        - CS, safety